SUMANTH R

Chennai, Tamil Nadu
B.Tech. in Computer Science and Engineering (AIML)
SRM Institute of Science and Technology

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Education

Degree	Institute	CGPA	Duration
B.Tech., Computer Science and Engineering (AIML)	SRM Institute of Science and Technology, Chennai, Tamil Nadu	8.98 / 10	*2023 – 2027*

Experience

- **Works Applications India Pvt. Ltd.** *Dec 2025 – Jan 2026*
  Software Engineering Intern On-site
  - Built a full-stack application using **Java, Spring Boot, MySQL, and React**, developing 5+ **REST APIs** for business workflows.
  - Optimized **SQL queries, database operations, and indexing**, reducing backend response time by **25%** during testing.
  - Performed **API integration, validation, error handling, and debugging** across backend services, frontend components, and database workflows.

- **Hyphiin** *Apr 2026 – Jun 2026*
  Software Development Engineering Intern Remote
  - Developed **5+ REST APIs** supporting application workflows, authentication, validation, business logic, and database operations.
  - Contributed to **3+ full-stack modules**, integrating backend services, frontend components, and database-driven workflows.
  - Resolved **15+ bugs and integration issues** through systematic debugging, testing, and root-cause analysis, improving application reliability.

Projects

- **Distributed Job Scheduler** *2026*
  Technologies: Python, FastAPI, PostgreSQL, Redis, Docker, AWS, Linux
  - Developed a distributed job scheduling service with **REST APIs, task queues, worker coordination, retries, and failure handling**.
  - Implemented persistent job tracking, **concurrency control, idempotent execution, and retry mechanisms** using PostgreSQL and Redis.
  - Containerized application services using **Docker** and implemented structured logging and error-handling workflows for troubleshooting and failure diagnosis.

- **Cloud-Based Multi-Tenant Platform** *2026*
  Technologies: Java, Spring Boot, React.js, PostgreSQL, MongoDB, Docker, AWS
  - Built a multi-tenant application with **REST APIs, role-based access, workflow management, and database-driven services**.
  - Designed and optimized **SQL queries, indexing, and database access** to support reliable application performance.
  - Containerized application components using **Docker** and integrated backend services with validation, exception handling, and database workflows.

- **Cloud Infrastructure Monitoring & Incident Management Platform** *2026*
  Technologies: Python, FastAPI, PostgreSQL, Docker, AWS, Linux
  - Built a service monitoring platform to track **application health, service availability, system events, and operational metrics** through REST APIs.
  - Implemented **health checks, structured logging, alert conditions, and incident tracking** to support proactive issue detection and troubleshooting.
  - Developed failure-handling workflows for **incident classification, root-cause investigation, and recovery tracking** across monitored services.

Publications

- **Deep Learning Based Brain Tumor Classification using Orange Data Analytics** *2026*
  IEEE International Conference (SCEECS 2026) View Paper
  - Developed and evaluated a deep learning pipeline using a **16,000+ MRI image dataset** aggregated from Kaggle, BRISC, and FigShare.
  - Applied **data preprocessing, PCA, augmentation, and stratified 10-fold cross-validation**, reducing feature dimensionality by **75%** while retaining **95% variance**.
  - Achieved **96%+ accuracy** and **0.997 AUC**, with a **7% improvement over baseline methods** and **8% improvement in MCC**.

Achievements

- Selected as a **Software Engineering Fellow at Headstarter AI**, working on real-world software engineering and AI projects.
- Secured **3rd Place** in the **CodeMavens 2025 Hackathon** organized by the ACM-W SRM Student Chapter.
- Published research on **deep learning-based brain tumor classification** at the **IEEE International Conference (SCEECS 2026)**.

Technical Skills

- **Languages**: Python, Java, C++, JavaScript, TypeScript, SQL, Bash
- **Cloud & DevOps**: AWS, Docker, CI/CD, Git, GitHub, Linux, Containerization
- **Cloud Operations**: Application Monitoring, Logging, Health Checks, Incident Troubleshooting, Error Analysis, Deployment Support
- **Backend Engineering**: Spring Boot, FastAPI, Node.js, REST APIs, API Design, Authentication, Authorization
- **Databases**: PostgreSQL, MySQL, MongoDB, Redis, SQL, Data Modeling, Query Optimization, Indexing
- **Systems & Networking**: Operating Systems, Computer Networks, TCP/IP, HTTP/HTTPS, DNS, Distributed Systems, Concurrency, Caching
- **Core CS**: Data Structures & Algorithms, OOP, DBMS, System Design, Scalability